

SUYOG HANMANT BARGULE

Agentic AI and Computer Vision Engineer

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 <https://suyogbargule.github.io/portfolio>

Profile Summary

- AI Engineer with 5 years of experience in Computer Vision and Generative AI, specializing in real-time model optimization, neural networks, and face recognition. Proven track record of deploying AI systems that improved performance metrics by up to 81%. Experienced in agentic AI, traffic surveillance, assistive technology, and automated pipelines, with a passion for building impactful and intelligent systems.
- Successfully built an automated lead generation system with effective email outreach, boosting engagement and conversions. Proven track record in developing intelligent systems across surveillance, traffic monitoring, and assistive tech domains. Passionate about solving real-world problems through impactful AI innovation.

Work Experience

Simplifi Dentistry Solutions India Pvt Ltd , Lucknow (Remote)– AI Engineer

-  Oct 2025-current
- Built an AI automation tool using RAG, LangFlow, Playwright, and LangChain.
 - It helps dental clinics collect patient insurance info, verify benefits, and manage claim processing automatically.

Futops Technologies, Pune – AI & Computer Vision Engineer

-  Oct 2022-Aug 2025
- Developed an agentic AI-powered lead generation system in n8n using Apollo AI, Afify Scraper, and LinkedIn/company data to automatically find, analyze, and qualify leads. Created personalized outreach and follow-up emails, cutting manual prospecting by 80% and driving higher conversions.
 - Developed a local chatbot using Langchain + DeepSeek integrated via Ollama (14B model) and CUDA, hosted on company website.
 - Designed a traffic surveillance system to detect helmet, triple-riding, and red-light violations, improving accuracy by 81%. Introduced best snapshot technique to boost object detection from 40% to 75%. (SPOT Awarded)
 - Built a real-time assistive system for visually impaired users on AWS with 82% accuracy, leveraging cloud-native AI architecture to ensure scalable performance.
 - Optimized face recognition system to 95% recognition and 90% detection with age, gender, and mask support at 30–35 ms inference time, enabling Edge AI deployment in low-latency environments.
 - Implemented panoramic 360° CCTV imaging and low-light stitching using OpenCV and GStreamer. (SPOT Awarded)

AI4SEE Pvt. Ltd, Bangalore – AI Engineer

-  Feb 2021-Sep 2022
- Built a Flutter-based rope jump counter using a TensorFlow Lite model for real-time analysis.
 - Developed a mobile app for direct farmer-to-retail sales; boosted sales by 20% with a regression model predicting prices five days in advance.

Skills

- Programming Languages :** Python, C++ .
- Technologies :** Docker, Linux, ZeroMQ, AWS (Rekognition, S3, EC2, SageMaker), Azure, OpenGL, GStreamer, Kafka, Kubernetes, ONNX Runtime, TensorRT, Playwright.
- Libraries & Frameworks :** PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, ONNX, OpenVINO, CUDA, NumPy, Pandas.
- Databases & APIs :** SQL, PostgreSQL, MongoDB, Qdrant, Pinecone, FAISS, FastAPI, REST.
- Specialties :** YOLO, CNN, GAN, LLaMA, DeepSeek, Vision Transformers, NLP, LangChain, RAG, Generative AI, Agentic AI, n8n, Hugging face, LangChain, LangGraph, LangSmith, LangFlow.
- Core Skills :** Object Detection, Image Segmentation, Performance Optimization, Image Processing, Model Deployment, Agile, Large Language Models (LLMs), Prompt Engineering, LoRA Fine-tuning, Multi-modal AI, Vector Databases, Stable Diffusion, Synthetic Data Generation.
- Soft Skills :** Teamwork and Collaboration,Adaptability,Analytical Thinking,Problem-Solving,Effective Communication.

Languages

English, Hindi, Marathi

Education

Master of Computer Application
Shivaji University Computer Science Department | Graduation Year: 2019

Certifications

Data Science and Machine Learning, ExcelR Solutions, Jan 2019 – May 2019
PyTorch for Deep Learning Bootcamp, Udemy – Andrei Neagoie & Daniel Bourke, Jun 2024 – May 2025